GPA & CGPA Calculator

PROJECT DOCUMENTATION | JAVA CONSOLE APPLICATION

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Language	Java
Application type	Console-based academic marks calculator
Runtime	Programiz Online Java Compiler / JDK 8+

1. Abstract

The GPA & CGPA Calculator is a Java console application that helps students estimate academic performance from subject marks. Users enter the number of semesters, subjects in each semester, subject names, maximum marks and marks obtained. The program calculates semester-wise percentage and estimated GPA, then combines all entered marks to report the overall percentage and estimated CGPA.

2. Problem Statement & Objectives

Manual calculation of academic results across multiple subjects and semesters can be time-consuming and prone to arithmetic mistakes. This project automates the process and provides a clear summary.

OBJECTIVES

- Calculate semester-wise GPA estimates and percentages.
- Calculate overall CGPA estimate, percentage, total marks and marks obtained.
- Accept user input for multiple semesters and subjects.
- Demonstrate Java input handling, loops, conditionals and arithmetic.

3. Technologies & Requirements

Technology: Java, Scanner class, loops, conditional statements and arithmetic operations.
Software: Programiz Online Java Compiler or another Java IDE; JDK 8 or later.
Hardware: Any computer capable of running a modern browser or Java environment.

4. Scope

The current version supports multiple semesters, multiple subjects per semester, subject names, maximum marks and marks obtained. It is a console application and does not store results after the run.

System Design & Calculation Method

How input is processed and how results are generated

5. Program Workflow

INPUT	PROCESS	OUTPUT
Semesters, subjects, names, maximum marks and obtained marks	Validate marks; total marks by semester and overall; apply formulas	Semester GPA estimate, semester percentage, total marks, overall percentage and CGPA estimate

6. Calculation Formulas

Semester percentage	(Semester marks obtained ÷ semester maximum marks) × 100
Estimated semester GPA	(Semester marks obtained ÷ semester maximum marks) × 10
Overall percentage	(Total marks obtained ÷ total maximum marks) × 100
Estimated CGPA	(Total marks obtained ÷ total maximum marks) × 10

Academic accuracy: These formulas provide a simple estimate by scaling percentage to 10. Official GPA/CGPA may require credit-weighted grade points and university-specific rules. Use the institution's published formula for official results.

7. Sample Test Case

Semester	Subject	Maximum	Obtained
1	English	100	85
1	Java	100	90
2	Mathematics	100	80
2	Python	100	95

Expected summary: Total maximum marks = 400; marks obtained = 350; overall percentage = 87.50%; estimated CGPA = 8.75/10. Semester 1 percentage/GPA estimate = 87.50% / 8.75; Semester 2 = 87.50% / 8.75.

8. Testing & Limitations

Testing: Valid marks should produce semester and overall summaries. Marks below zero or above the maximum are rejected by the current version.
Limitations: The current version is console-based, does not save results, and uses a simplified percentage-to-10-point estimate rather than a university-specific GPA system.

9. Conclusion & Future Enhancements

This project demonstrates how Java can automate basic academic calculations. Future improvements may include university-specific formulas, improved input validation, saved student records, a graphical interface, and PDF report export.

10. References

Java documentation: https://docs.oracle.com/en/java/
Programiz Java tutorial: https://www.programiz.com/java-programming